Deploy Education App to Kubernetes

DevOps

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Project Number:DO-24

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1. Problem Statement

In modern education, learners often struggle to understand complex concepts due to a lack of engaging, structured, and visually enriched study material. Traditional learning resources are mostly text-heavy, static, and do not adapt to individual learning needs.

At the same time, generating high-quality educational content manually is time-consuming for educators. There is a need for a system that can automatically generate **structured, interactive, and visually supported learning content** from simple topic inputs.

Additionally, deploying such AI-powered applications in a scalable and reliable way is challenging without proper DevOps practices.

Therefore, the problem is to design and deploy a system that:

- Automatically generates educational content using AI
- Enhances learning with visuals and structured explanations
- Ensures scalability, reliability, and real-world usability through DevOps practices

2. Project Objectives

The main objectives of this project are:

Primary Objectives

- To develop an **AI-powered education application** that generates structured content from a given topic
- To integrate **text generation (LLMs)** and **image generation APIs** for better understanding
- To create an **interactive user interface** using Streamlit

Technical Objectives

- To implement a **backend system** that processes user input and interacts with AI services
- To containerize the application using **Docker**
- To deploy the application on **cloud infrastructure (AWS EC2)**
- To implement **CI/CD pipelines (GitHub Actions)** for automated deployment

Performance & Usability Objectives

- To ensure **scalability and high availability**
- To provide **real-time content generation**
- To make the system **user-friendly for students and educators**

3. Scope of the Project

The scope of this project includes:

In Scope

- Development of an AI-based educational content generator
- Integration of:
    - Text generation APIs (e.g., Gemini / LLMs)
    - Image generation APIs (e.g., Hugging Face)
- Web-based interface using Streamlit
- Deployment using:
    - Docker containers
    - AWS EC2 / Kubernetes
    - CI/CD pipelines using GitHub Actions
- End-to-end DevOps lifecycle implementation

Out of Scope

- Offline functionality (requires internet for API calls)
- Advanced personalization based on user learning history
- Mobile application development
- Full-scale Learning Management System (LMS) features

Target Users

- Students for self-learning
- Educators for content creation
- Developers/DevOps engineers for learning deployment practices

Proposed Solution

1. Key Features

- AI-powered content generation using Large Language Models (LLMs)
- Structured concept explanation for better understanding
- Key learning points extraction
- AI-generated visuals for concept visualization
- Real-time topic-based content generation
- Interactive web interface using Streamlit
- Cloud deployment for global accessibility
- Containerized application using Docker
- Automated CI/CD pipeline using GitHub Actions
- Scalable and production-ready architecture

2. Overall Architecture / Workflow

The system follows a modular and cloud-based architecture that integrates AI services with a user-friendly interface.

Workflow:

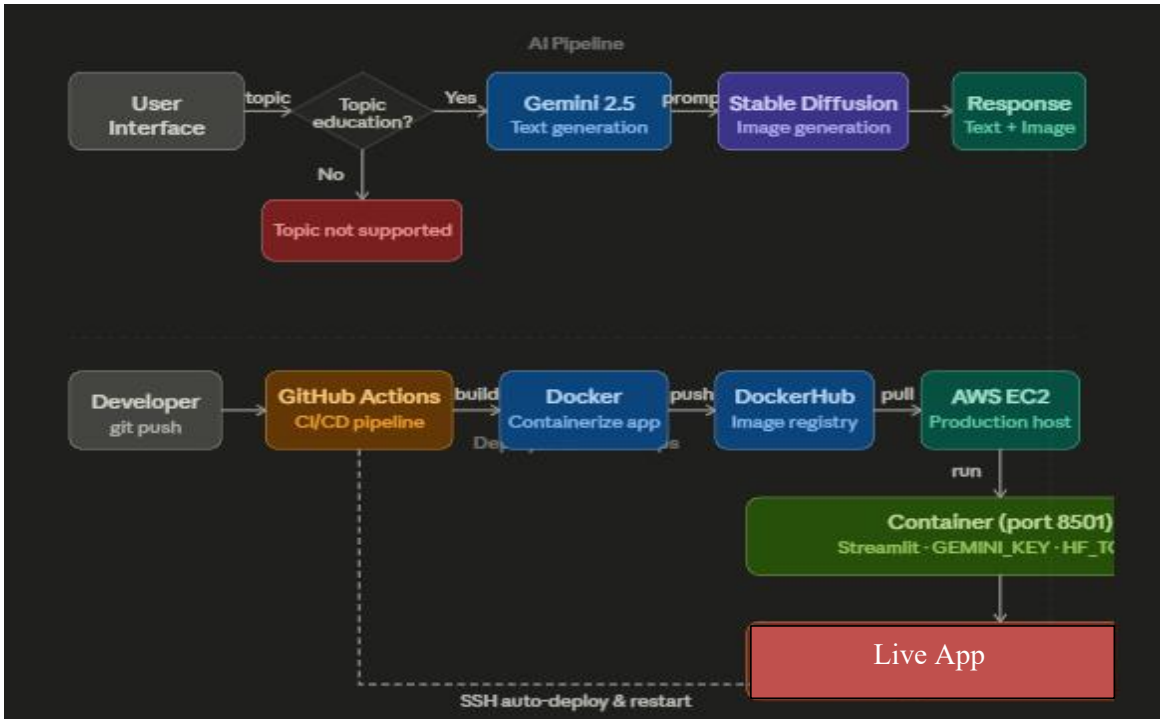
1. User enters a topic through the Streamlit web interface
2. The request is sent to the Gemini API for text generation
3. Generated content is processed into structured educational format
4. A prompt is created for visual generation
5. Hugging Face API generates relevant images
6. Text and images are combined and displayed to the user

Deployment Workflow:

- Code is pushed to GitHub repository
- GitHub Actions triggers CI/CD pipeline
- Docker image is built and pushed to DockerHub
- Application is deployed on AWS EC2
- Container is automatically updated and restarted

User Input → Text Generation → Processing → Image Generation → UI Display

Application Design



3. Tools & Technologies Used

AI & Processing Layer

- Gemini API (Text Generation)
- Hugging Face (FLUX / Stable Diffusion for Image Generation)

Application Layer

- Python (Backend logic)
- Streamlit (Frontend UI)

DevOps & Deployment

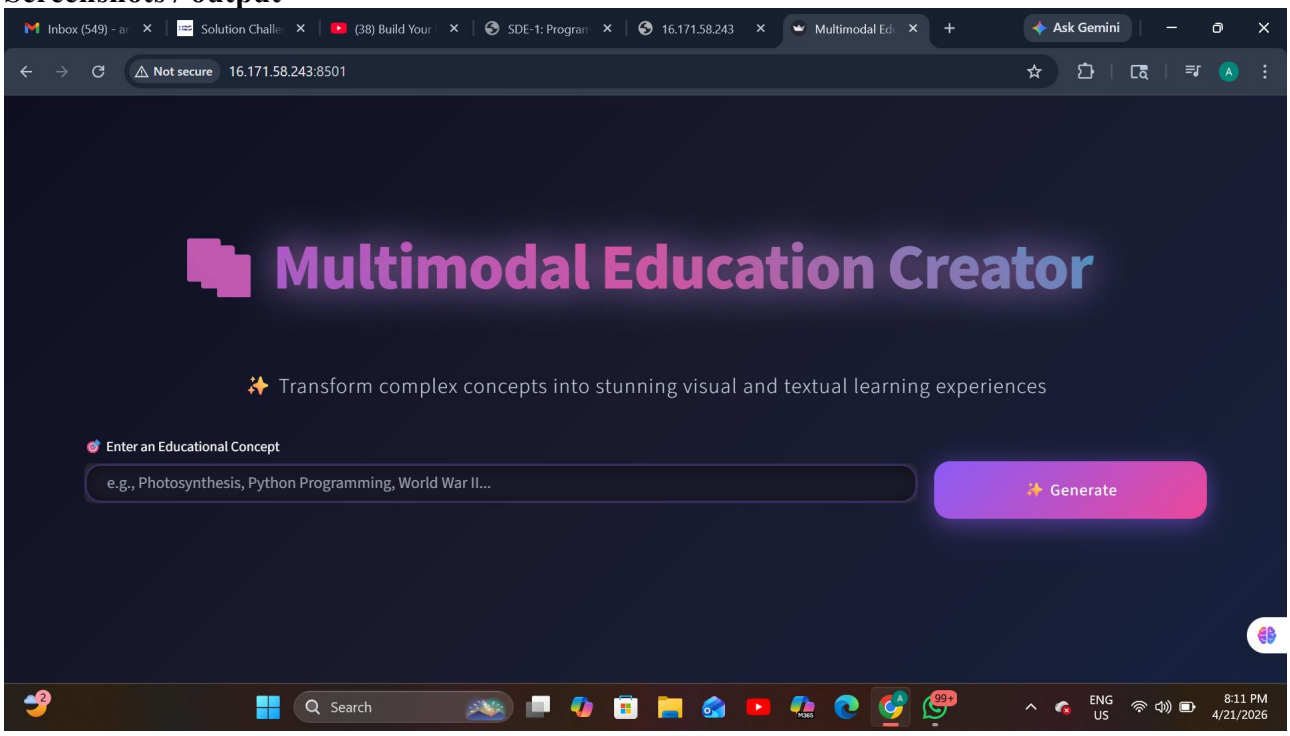
- Docker (Containerization)
- AWS EC2 (Cloud Hosting)
- GitHub Actions (CI/CD Automation)

Other Tools

- Git & GitHub (Version Control)
- Environment Variables (.env) for secure configuration

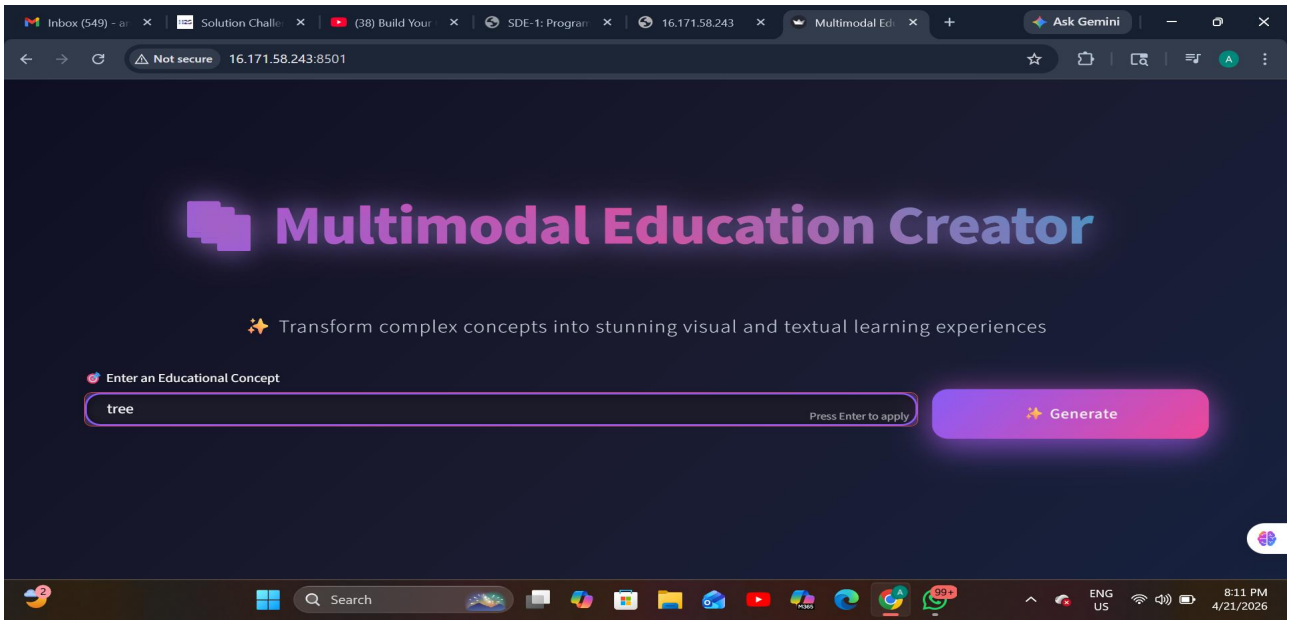
Results & Output

Screenshots / output



Live Application Details:

- **Hosting Platform:** AWS EC2
- **Public IP:** 16.171.58.243
- **Port:** 8501
- **Access URL:** http://16.171.58.243:8501



Conclusion

The **Multimodal Education Creator** successfully demonstrates the integration of Artificial Intelligence and DevOps practices to build a real-world, scalable application. The project addresses the challenge of generating engaging and structured educational content by combining text generation using LLMs and AI-based image generation.

Through this project, we implemented a complete end-to-end workflow—from idea development to deployment—using modern technologies such as Streamlit, Docker, AWS EC2, and CI/CD pipelines. The system is capable of generating real-time, visually enriched learning material from simple topic inputs, making it highly useful for students and educators.

Key Learnings:

- Understanding and integrating Large Language Models (LLMs) in applications
- Working with AI APIs for both text and image generation
- Building interactive web applications using Streamlit
- Implementing containerization using Docker
- Deploying applications on cloud platforms (AWS EC2)
- Automating workflows using CI/CD pipelines (GitHub Actions)
- Gaining hands-on experience with real-world DevOps lifecycle

Overall, the project highlights how AI and DevOps together can create scalable, efficient, and impactful solutions in the education domain.

Future Scope & Enhancements

The project can be further enhanced with the following improvements:

- **User Authentication & Personalization**
  Add login/signup functionality and personalized learning experiences
- **Database Integration**
  Store user history, generated content, and preferences for future access
- **Multi-Model Support**
  Integrate additional AI models for improved accuracy and variety
- **Advanced UI/UX Improvements**
  Enhance the interface for better user experience and responsiveness
- **Mobile Application Development**
  Extend the platform to Android/iOS applications
- **Offline Support (Partial)**
  Cache previously generated content for offline access
- **Performance Optimization**
  Improve response time and system efficiency
- **Kubernetes Deployment**
  Upgrade from EC2 to Kubernetes for better scalability and orchestration

- **Multilingual Support**
  Generate content in multiple languages for wider accessibility
- **Integration with LMS Platforms**
  Connect with existing learning systems like Moodle or Google Classroom

GitHub: https://github.com/nameisankit/education-app